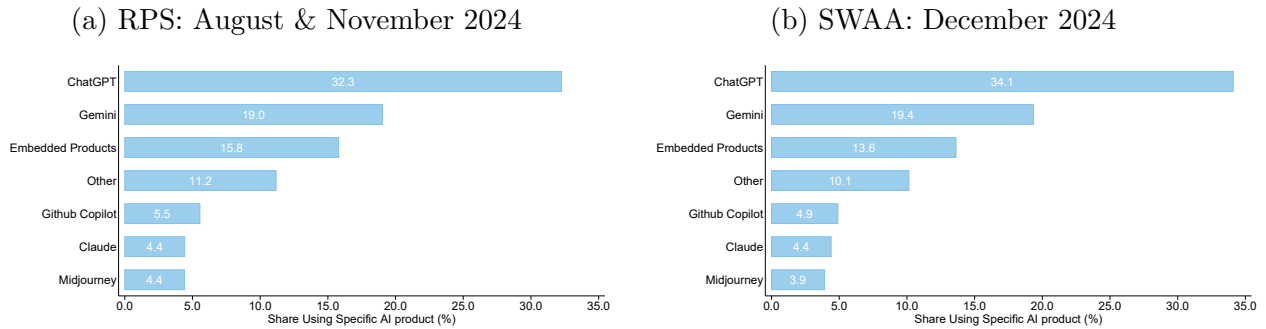


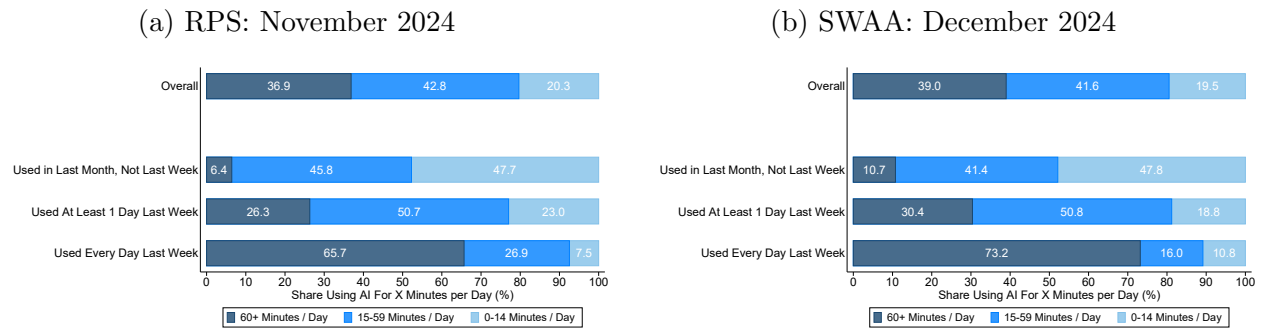
### C.3 Comparison of Generative AI Use in the RPS and SWAA

FIGURE C.2: Share of Respondents Using Specific Generative AI Products



*Notes:* The figure shows the share of respondents who report using particular genAI products. “Embedded products” are genAI features embedded within existing software, such as Microsoft Copilot. Data source is the August and November 2024 waves of the RPS, ages 18-64 ( $N = 9742$ ) and the December 2024 SWAA, ages 20-64 ( $N = 4698$ ). Individuals who report using multiple genAI products are reflected in multiple bars.

FIGURE C.3: Intensity of genAI Use For Work



*Notes:* The figure shows the distribution of daily time spent actively using genAI for work, among genAI users. Usage time is broken down into 0-14 minutes per day (light blue), 15-59 minutes per day (medium blue), and 60 or more minutes per day (dark blue). The “Overall” bar reflects the distribution among all genAI users. In Panel a, data source is the November 2024 waves of the RPS, ages 18-64. In Panel b, data source is the December 2024 SWAA, ages 20-64. The sample for both data sources is employed respondents who use genAI for work. (RPS,  $N = 1000$ ; SWAA,  $N = 861$ ).

## D Details on Time Savings and Productivity Calculations

### D.1 Derivations Using the Aggregate Production Model

#### The Aggregate Productivity Gain from Generative AI

We model the economy's aggregate output using a Cobb-Douglas aggregate production function where labor supply is perfectly substitutable across workers. The economy contains  $N$  workers. Let  $Y$  denote the aggregate output, and let  $L$  denote the aggregate supply of labor measured in efficiency units (going forward we refer to this as aggregate effective labor supply). The aggregate production function is given by:

$$Y = AK^\alpha L^{1-\alpha}, \quad (\text{D.1})$$

where  $A$  is TFP,  $K$  is the aggregate capital stock, and  $\alpha$  is the Cobb-Douglas share on capital. The aggregate effective supply of labor  $L$  is defined as the weighted sum of individual labor supplies:

$$L = \sum_{i=1}^N \ell_i e_i \quad (\text{D.2})$$

where  $\ell_i$  is the number of hours worked per week by worker  $i$ , and  $e_i$  is the efficiency units of labor supplied by worker  $i$ .

In a competitive labor market, each worker is paid their marginal product of labor:

$$w_i = (1 - \alpha)A(K/L)^\alpha e_i \quad (\text{D.3})$$

where  $w_i$  is the worker's hourly wage. Normalizing the mean efficiency unit to  $\bar{e} = 1$ , we have:

$$\tilde{w}_i \equiv \frac{w_i}{\bar{w}} = \frac{w_i}{(\sum_i w_i)/N} = \frac{(1 - \alpha)A(K/L)^\alpha e_i}{(\sum_i (1 - \alpha)A(K/L)^\alpha e_i)/N} = \frac{(1 - \alpha)A(K/L)^\alpha e_i}{(1 - \alpha)A(K/L)^\alpha \bar{e}} = e_i \quad (\text{D.4})$$

where  $\tilde{w}_i$  is worker  $i$ 's wage relative to mean wages  $\bar{w}$ .

Suppose that worker  $i$  saves  $s_i$  hours per week due to genAI and spends this time on additional production within their job. The effective weekly labor supply for worker  $i$  is then  $\ell_i + s_i$ . Substituting this into the expression for aggregate effective labor supply, we have:

$$L' = \sum_i (\ell_i + s_i) e_i = \sum_i \ell_i e_i + \sum_i s_i e_i. \quad (\text{D.5})$$

The change in aggregate effective labor supply attributable to genAI-induced time savings

is:

$$\Delta L \equiv L' - L = \sum_i s_i e_i = \sum_i s_i \tilde{w}_i \quad (\text{D.6})$$

Assuming no change in TFP, capital, and hours worked, the percent change in aggregate productivity from genAI equals the percent change in aggregate output. The approximate percent change in aggregate output from genAI is given by:

$$\frac{\Delta Y}{Y} \approx (1 - \alpha) \frac{\Delta L}{L} = \underbrace{(1 - \alpha)}_{\text{labor cost share}} \times \underbrace{\frac{\sum_i s_i \tilde{w}_i}{\sum_i \ell_i \tilde{w}_i}}_{\Delta \% \text{effective labor}} \quad (\text{D.7})$$

This equation states that the percent change in output due to genAI is the ratio of mean time savings to mean hours worked, weighted by worker's wages, and scaled by labor's share of production costs.

Using November 2024 RPS data on hourly wages  $w_i$ , weekly hours worked  $\ell_i$ , and weekly genAI time savings  $s_i$ , we use (D.7) to estimate that genAI currently increases aggregate effective labor supply by 2.1%. If we assume an AI-exposure-adjusted labor share of 0.57, following Acemoglu (2024), the implied potential productivity gain is 1.2%.

### Comparisons to Micro and Macro Estimates of GenAI Productivity Gains

How does our estimate of the aggregate productivity gain from genAI compare with experimental estimates from the literature? To answer this question, we reformulate the model to express time savings as a linear function of time spent using the technology:

$$L' = \sum_i (l_i + \gamma u_i) e_i = \sum_i l_i e_i + \gamma \sum_i u_i e_i \quad (\text{D.8})$$

In this expression,  $u_i$  is the weekly hours spent using genAI by worker  $i$ , and  $\gamma$  is the productivity gain associated with one hour of genAI use.

We can then write the percent increase in aggregate effective labor supply due to genAI as

$$\frac{\Delta L}{L} = \frac{\gamma \sum_i \tilde{w}_i u_i}{\sum_i \tilde{w}_i l_i} \quad (\text{D.9})$$

Next, we solve for the value of  $\gamma$  that would generate a 2.1 percent increase in aggregate effective labor supply given our estimates of genAI use:

$$\gamma = 2.1\% \cdot \left( \frac{\sum_i \tilde{w}_i l_i}{\sum_i \tilde{w}_i u_i} \right) \quad (\text{D.10})$$

Recall that for each worker in the RPS data we do not have a single estimate for  $l_i$ ; instead, we have a lower and upper bound,  $\underline{l}_i$  and  $\bar{l}_i$ , respectively. Evaluating (D.10) using the midpoint of these bounds,  $l_i \equiv (\underline{l}_i + \bar{l}_i)/2$ , yields a wage-weighted share of total work hours spent using genAI of 7.0%, which implies a productivity parameter value of  $\gamma = 0.30$ . This implies that each hour spent using genAI increases the worker’s productivity for that hour by 30%. This is similar in magnitude to the average value of 27% from several randomized experiments of genAI usage (Cui et al., 2024; Dell’Acqua et al., 2023; Noy and Zhang, 2023; Peng et al., 2023).